

Intermediate abundance promotes speciation when dispersal is limited

Abstract

Why do some lineages diversify while others do not? This remains a central question in evolutionary ecology. A long-standing assumption, dating to Darwin and embedded in the Unified Neutral Theory of Biodiversity, holds that abundant species should speciate at higher rates. Conversely, theoretical and empirical work highlights the possibility that rare and dispersal-limited clades might be more prone to speciation. Using a birth-death-immigration model with protracted speciation in a multi-population landscape connected by limited dispersal, we show that abundance has a hump-shaped effect on probability of speciation. Our model reveals that intermediate abundance maximizes speciation probability because larger populations disperse more, swamping regional differentiation and inhibiting speciation completion, while smaller populations lack the persistence and incipient speciation needed to diversify. We find empirical support for this prediction with an analysis of data from arthropods endemic to Hawai'i, where genus-level species richness shows a significant hump-shaped relationship with mean genus abundance. These findings provide a mechanistic explanation for a nuanced relationship between abundance and diversification.

19 Introduction

20 A fundamental question in evolutionary ecology concerns why some clades are diverse while
21 others are not. Are there properties of species and lineages that promote speciation? One line of
22 inquiry with a long and convoluted history is the role of commonness and rarity in the process of
23 diversification. Going back over 150 years, Darwin (1859) argued that widespread, abundant
24 species should be superior competitors and thus more likely to give rise to new species.

25 Building on this foundation, Yule's (1925) model, which became a landmark contribution to
26 modeling diversification, was initially intended as a way to incorporate demography into
27 predictions for diversification and clade richness (Pennell and MacPherson 2025). Despite these
28 deep roots in evolutionary thinking, little consensus has emerged on the relationship between
29 abundance and diversification. Darwin's line of thinking carried forward to macroecologists who
30 attempted to model, with some empirical support, the presumed positive association between
31 commonness, fitness, and speciation (Maurer 1999).

32 In parallel, paleontologists searched for relationships between commonness and diversification,
33 alternately finding evidence for a positive relationship (Krug et al. 2008) but also a negative
34 relationship (Stanley 1986, Jablonski and Roy 2003). Empirical studies of modern taxa have
35 largely found that commonness is negatively or inconclusively related to measures of
36 diversification (Jablonski and Roy 2003, Makarieva and Gorshkov 2004, Greenberg and Mooers
37 2017, Smyčka et al. 2023, Afonso Silva et al. 2025, but see Hay et al. 2022). While some of
38 these studies focus specifically on average abundance, others on range size, and some on both,
39 the two macroecological properties are most often strongly correlated, a fascinating
40 macroecology pattern in and of itself (Brown 1995, Gaston 2003). This correlation may be
41 caused, at least partly, by the net increase in dispersal of larger populations owing simply to
42 chance: if each individual carries a probability of dispersing, more individuals in a population
43 will lead to more overall dispersal (Hubbell 2001). It is also of course possible that shared
44 underlying mechanisms cause both increased population growth rate and increased dispersal rate
45 (Gaston et al. 1997, Holt et al. 1997).

46 More recent theoretical and modeling work has largely assumed a positive relationship between
47 commonness and diversification. Hubbell (2001) perhaps started this trend with the inclusion of
48 "point mutation" speciation in the Unified Neutral Theory of Biodiversity (UNTB). Because
49 point mutation speciation assumes a constant probability of speciation for every individual,
50 lineages with more individuals will experience more speciation events in the UNTB (Hubbell
51 2001, Etienne et al. 2007). The speciation mechanism first proposed by Hubbell (2001) was
52 never an accurate model of real diversification (Ricklefs 2003), but even later attempts to
53 increase the realism of speciation in the UNTB retain the emergent property that more abundant
54 lineages will undergo more speciation. Protracted speciation (Rosindell et al. 2010) again
55 assumes that incipient speciation is constant across individuals and thus full speciation will scale
56 positively with lineage abundance; fission speciation (Etienne and Haegeman 2011) also
57 assumes the probability of fission (the event that leads to speciation) increases with the number
58 of individuals in a species. Largely independently from UNTB and its descendants, phylogenetic
59 models of geographic change and diversification also explicitly assume that more widespread
60 species have greater opportunity for speciation (Goldberg et al. 2011).

The process that underlies speciation itself may suggest something different in terms of the relationship between speciation, commonness, and rarity: dispersal. Dispersal is a key mechanism by which populations can become isolated or connected, potentially leading to speciation or admixture (Yamaguchi 2022). Dispersal figures prominently in phylogenetic models of the geographic range and diversification (Matzke 2014), it is necessary to maintain local biodiversity in the UNTB (Hubbell 2001), and it is the mechanism connecting abundance to range size in both neutral (Hubbell 2001) and non-neutral (Brown 1995) macroecological models.

Due to the potential for dispersal to both create population isolates but also swamp out regional differences that could have led to speciation, the connection between dispersal ability as a biological trait and speciation as an evolutionary outcome remains a fertile line of inquiry. Some empirical studies have found that measures of diversification have a hump-shaped or negative relationship with morphological proxies for dispersal ability (Price and Wagner 2004, Claramunt et al. 2012, Czekanski-Moir and Rundell 2019) while others have found a positive to flat relationship between dispersal ability and diversification (Claramunt et al. 2025). Agent-based model simulations have supported the idea of a trade-off between too little dispersal leading to population instability and too much dispersal leading to admixture with intermediate dispersal balancing the two extremes allowing for speciation (Birand et al. 2012, Ashby et al. 2020, Ciccheto et al. 2024).

But dispersal is not just a trait connected to morphology or imposed in a simulation, it emerges from population dynamics: each individual carries some per-capita probability of dispersing, but because that probability is applied across every individual in the population, the total dispersal rate (the total number of dispersal events per unit time) scales linearly with abundance, thus species with more individuals produce more dispersal overall. This is the assumption, a realistic one, in birth-death-immigration models (Kendall 1948). Therefore, if commonness and dispersal affect speciation, their effects are interconnected because population-level dispersal rate increases both with intrinsic per-capita dispersal probability and with abundance. It also remains an open debate whether their effects result from the determinism of adaptive evolution as Darwin (1859) believed or instead emerge from chance.

Here we develop a birth-death-immigration model with protracted speciation embedded in a landscape of multiple local populations connected with limited dispersal to investigate the role of abundance and stochastic dispersal in modulating the probability of speciation. Contrary to Darwin (1859) and UNTB (Hubbell 2001, Etienne et al. 2007, Rosindell et al. 2010, Etienne and Haegeman 2011), we find that intermediate abundances lead to the greatest probability for speciation. Critically, this result does not depend in any way on whether populations are more or less adaptively fit, it only depends on a balance between large enough abundances to accrue sufficient probability of incipient speciation but small enough abundances to not lose regional differentiation due to increased dispersal. We also analyze real data on the species richness and abundance of arthropod communities in Hawai'i finding empirical evidence for intermediate abundance promoting speciation. The isolation of Hawai'i, leading to tractable endemic radiations, and its known geologic age, ranging from still forming to 4–6 million years, make it an ideal study system for investigating the interplay of abundance and speciation.

Methods

Simulating a birth-death-immigration process with protracted speciation

We simulate a birth-death-immigration process with protracted speciation in a modified metapopulation (Hanski 1998) setting [Figure 1](#). For simplicity, we consider only a single species (not a multi-species) metapopulation. Constraining our simulations to a single species allows us to focus on the processes of interest—population dynamics and speciation—without entering into debates on multi-species dynamics (HilleRisLambers et al. 2012, Rosindell et al. 2012).

In this simulation framework there are a number of local populations (minimum two are needed, we simulate 10) fully connected by local dispersal as well as a global source population connected to each local population by dispersal. The global source population is considered extremely large relative to local populations such that extinction of the global source population is unlikely on the time scale of local dynamics (we approximate this assumption by never allowing the source population to go extinct). Local populations start at size 0 and grow and shrink according to local births and deaths, dispersal between local populations, and dispersal from the source pool.

Our simulation model is consistent with other simulation models derived from the UNTB (Hubbell 2001, Rosindell and Harmon 2013, Rosindell et al. 2015), but unlike the multi-species perspective of those models, we concern ourselves only with a single species and whether that one species undergoes a speciation event. In this way we do not make the assumption of per capita neutrality across species, meaning our results could generalize to multi-species eco-evolutionary models based on niche theories (e.g. Tilman 1982, Chesson 2000, Letten et al. 2017). We also relax the zero-sum and fixed time step approaches of most UNTB-like models. Instead, we initialize all patches as empty and let their numbers of individuals grow and shrink with birth, death, and dispersal. We also use a Gillespie algorithm (Gillespie 1976, 1977) to simulate dynamics. In the Gillespie algorithm each discrete time step is an event chosen at random in proportion to the rate of occurrence of that event (i.e. not the fixed death-replacement cycle of the UNTB; Hubbell (2001)). Although events are sampled at discrete time steps, time itself is measured in actual units of time, not time steps, by sampling sojourn times between events from an exponential distribution with rate equal to the sum of all rates for all possible events.

Protracted speciation

In our simulation model a new species arises via a modified protracted speciation process (Rosindell et al. 2010) similar to the model of Tao et al. (2021). Protracted speciation was first proposed by Rosindell et al. (2010) to try to bring more realism to the speciation process of the UNTB which originally modeled speciation as a point mutation-like process in which a single individual gets randomly flipped to a new species identity with some small probability ν (Hubbell 2001). Protracted speciation instead assumes that this point mutation-like process, again with probability ν , only flips an individual to be an *incipient* species; if the lineage of that incipient species survives τ time steps, the process is said to reach full speciation but if the lineage dies out before τ time steps, no new species is formed (Rosindell et al. 2010).

We extend the protracted speciation model to a multi-patch simulation landscape. In our model, after a population is founded through immigration it, the newly founded population, has a small probability of becoming an incipient new species. If an incipient speciation event occurs, and in the absence of subsequent immigration, the population of the incipient new species must persist τ time units for full speciation. If the population does not persist for τ time units, the incipient new species is lost without ever becoming a fully new species.

After an incipient speciation event, subsequent immigration events slow the progress toward full speciation. We model this by adding a small increment to τ equal to ξ/n_i where ξ is a parameter we can set and n_i is the size of the population containing the incipient species. Due to n_i in the denominator, this slowdown will be felt less by large populations and more by smaller populations in agreement with population genetic theory on the effect of immigration on population differentiation (Wright 1931, Mills and Allendorf 1996).

Tao et al. (2021) also developed a similar multi-patch protracted speciation model, but ours is different in three ways:

1. Tao et al. (2021) assume every population founded by immigration is an incipient species; we assume that a new population founded through immigration remains part of the parent species unless an explicit incipient speciation event occurs.
2. In our Gillespie algorithm approach, time to full speciation is measured in actual time units, not discrete time steps.
3. Tao et al. (2021) assume that a fixed constant is added to τ for every immigration event while we add a constant scaled by local population size (ξ/n_i).

Simulation rules

The preceding sections explained our simulation model, summarized in [Figure 1](#); here we summarize the rules by which the simulation progresses:

1. All local populations are initialized as empty, to be filled by immigration.
2. Per capita birth, death, immigration, and incipient speciation all happen independently and are determined by their own respective rates.
3. Immigration is modeled as the dispersal of a newly born individual, thus the population from which the individual emigrates is unchanged; this birth-then-emigrate mechanism is a widely used simplification in models inspired by the UNTB (Hubbell 2001, Rosindell and Harmon 2013, Rosindell et al. 2015, Tao et al. 2021). Each individual carries a per capita rate of producing a dispersal event, thus emigration from a patch scales with the population size of that patch, but dispersal from the source pool is not density dependent. The recipient population of an immigration event is chosen uniformly at random, the population size of the receiving population does not impact probability of receiving an immigration event.
4. Speciation has 2 steps:
 - i. After a population is founded through immigration, incipient speciation can change that population to an incipient new species with a per capita rate of ν . Subsequent incipient speciation events in the same local population are ignored because the population is already an incipient new species. Multiple incipient species can occur across different patches, but not within the same patch.

- ii. If an incipient species survives long enough it becomes a new species; “long enough” is determined by a waiting time parameter τ —the larger τ , the longer the time frame for a fully new species to emerge.
5. Immigration between local communities and from the global source pool slows the progress toward full speciation; if an incipient species has to wait τ time units (in the absence of immigration) until it is a full species, each immigration event adds an increment to τ of ξ/n_i .
6. If an individual of an incipient species disperses to a different population, it is subsumed within that population and its incipient status is lost.
7. If a local population goes extinct, the speciation tracking for that patch is reset.
8. The simulation is stopped under one of two conditions: i) if any incipient species becomes a fully diverged new species, or ii) if the simulation reaches the maximum designated number of iterations.

Because of the stopping conditions, each simulation replicate will either yield one and only one full speciation, or no full speciation events at all.

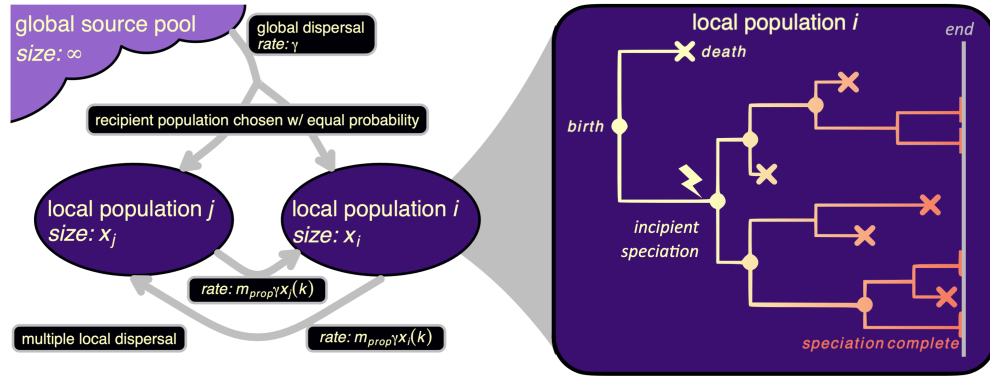


Figure 1: Conceptual overview of our simulation model. Local populations are connected by immigration between populations and also from a global source pool. Within local populations birth and death determine population growth and decline, augmented by immigration. Random incipient speciation events can occur at the local population level but do not lead to full speciation unless the population persists for at least a duration of τ (possibly more in the face of immigration). If full speciation occurs the simulation stops. If no speciation occurs but the maximum number of iterations is reached, the simulation likewise stops.

Further details of the simulation are provided in Supplementary Section S2. All simulation code is written in R (R Core Team 2025) using *Rcpp* (Eddelbuettel et al. 2025a) and *RcppArmadillo* (Eddelbuettel et al. 2025b). Underlying C++ and R source code is available in the R package *abundolism* accompanying this paper. Supplementary Section S1 explains how to access this custom R package.

Simulation experiment

Table 1 summarizes the simulation parameters in our model and the range of values we use in our simulation experiment. In all cases parameter values are drawn from a uniform distribution.

A total of 5,000 simulations were run, each with a unique set of randomly sampled parameters. Parameter values and the number of iterations were chosen with three considerations in mind: 1) speciation should be possible to observe in a reasonable number of simulation iterations for computational practicality; 2) abundances should be within realistic ranges; and 3) timescales should be within reasonable bounds.

In the context of the speciation-related parameter values we used, timescales should be interpreted in units of 10,000 years. Thus the maximum value of τ is 2,000,000 years and the maximum rate of incipient speciation ν is 0.1 per individual per 10,000 years, leading to rates of new species divergence roughly in line with published estimates (Hedges et al. 2015).

Table 1: Parameters governing the simulation and their ranges over the simulation experiment. Mathematical parameter names are given first followed by their name in the simulation code. Note n_p and n_{step} were fixed throughout the course of the simulation experiment.

parameter	range	description
λ (la)	0.1 4.0	per capita local birth rate
μ (mu)	0.1 4.0	per capita local death rate
γ (g)	0.1 1.0	dispersal rate from the global source pool to one local population
m_{prop} (m_prop)	0.0 0.1	proportional dispersal rate between local populations relative to global rate
ν (nu)	0.0 0.1	per capita incipient speciation rate
τ (tau)	20.0 200.0	wait time to full speciation in the absence of dispersal
ξ (xi)	10.0 100.0	amount each dispersal event sets back the progression toward speciation
n_p (np)	10.0 10.0	number of local populations
n_{step} (nstep)	10,000.0 10,000.0	number of simulation iterations

Parameter values were chosen as a balance between biological realism and computational efficiency: rates are sufficiently fast to allow for abundances to reach levels often found in survey data (e.g. the arthropod survey data we analyze here) within 10,000 time steps.

Analysis of simulation experiment results

Our primary focus is on the relationship between abundance and probability of speciation. We do this by comparing the presence or absence of a full speciation event to species abundance averaged over time and patches. However, we must account for the stopping conditions of our simulation being dependent on the very process—speciation—that we are analyzing. Simulation runs reaching full speciation will have fewer time steps than those reaching the pre-determined time step limit. We account for this by cutting each simulation run into a set of time step windows of fixed width (i.e. fixed number of time steps). We then analyze the presence or absence of speciation and average abundance within these non-overlapping windows of fixed width. A window is marked as containing a speciation event or not. We calculate abundance as

the average population size across time steps of the entire metapopulation. We apply this approach to the analysis of speciation versus abundance, but also to auxiliary analyses of how speciation and abundance respond to model parameters. Those auxiliary analyses include visualizing average abundance across different birth and death rates and dispersal rates, as well as fitting models of the probability of speciation as a function of model parameters. Because choice of window size could affect analysis results, we re-run the analysis of abundance versus speciation using windows of different width.

In fitting any models with these simulation results cut into fixed-width windows, we must account for the non-independence between windows from the same model run. We do this using generalized linear mixed effects models [using R package *glmmTMB*; McGillicuddy et al. (2025)] where simulation run is a random effect. We report confidence intervals rather than hypothesis tests because sample sizes from simulations are arbitrary and thus degrees of freedom can be manipulated to produce statistically significant results from biologically meaningless patterns. By contrast, confidence intervals more meaningfully indicate the magnitude of effects in the simulation experiment.

Analysis of arthropod richness and abundance from Hawai`i

To bring real data to bear on our insights from simulations, we analyze the species richness and average per-species abundance of arthropod genera in Hawai`i. Endemic arthropods in Hawai`i are a uniquely ideal study system because, although only an approximation, most arthropod genera only arrived in Hawai`i once, meaning all endemic species of those genera represent *in situ* diversification (Cowie and Holland 2008). For the purpose of our analysis, we proceed with this approximation. The unique situation in Hawai`i, that each arthropod genus arrived approximately only once on this geologically young archipelago, also avoids the problem that taxonomic ranks are temporally inconsistent at a global level (Avise and Liu 2011).

To calculate species richness per genus we use the Bernice Pauahi Bishop Museum's checklist of arthropods in Hawai`i (Nishida 2002). This checklist represents the current understanding of terrestrial arthropod species richness in Hawai`i, though is inevitably incomplete for a place supporting rapid diversification. To estimate average per-species abundance for each genus we use an extensive survey of arthropods from across the Hawaiian Archipelago (Gruner 2007). The survey data from Gruner (2007) comprise sample locations from 45 trees across 5 sites. Sites are located on Hawai`i Island, Moloka`i, and Kaua`i and follow the Long Substrate Age Gradient plots established by Vitousek et al. (1995) that are constrained to be within similar montane rainforest environments.

Combining the checklist with survey data resulted in the retention of 118 endemic genera out of 597 total from the checklist, representing a loss of 80.23%. While this is a large percentage, it is also not unexpected. The survey data constitute a very small sample of the geographic space of the Hawaiian Archipelago as well as a small sample of the extremely heterogeneous environmental space of the archipelago. Supplementary Section S5.4 evaluates the impact of these missing genera on our analyses, finding that our results are robust to this subsampling of total genera.

Average per-species abundance was estimated for each genus as the mean of each species's abundance for all species belonging to that genus across all sample locations. We use average

per-species abundance to represent the measure of commonness or rarity of a genus because speciation occurs at the species level, or lower, so abundance of higher taxonomic groups is not as relevant. We can also estimate the average abundance of a species with lower error than the total abundance of a genus. In Supplementary Section S5.5 we further study the choice of using average per-species abundance and find it to be a rigorous one.

We are interested in determining if the number of species per genus (a proxy for diversification) peaks at intermediate per-species abundances. To do this we fit a quadratic generalized linear model with species richness as response and average per-species abundance as explanatory variable. But the question remains, what probability distribution should we assume for number of species? Kendall (1948) showed that a birth-death process will result in a geometric distribution of descendants. Conditional on lineage survival at time t after initial establishment, we can parameterize the geometric probability mass function as

where y is the number of species in a lineage. $\psi(t)$ is the probability that a lineage survives but we do not observe any bifurcation events by time t . Thus $1 - \psi(t)$ is the probability of the complement which is observing at least one bifurcation event leaving multiple descendants at time t . We will call $1 - \psi(t)$ the “observable diversification probability.” $\psi(t)$ depends on rates of speciation and extinction in addition to time t .

The number of species in any given lineage i extant at time t will follow a geometric distribution with probability parameter $p_i = \psi_i(t_i)$. But we do not know p_i because we do not know the arrival times of most arthropod lineages in Hawai‘i—thus we do not know t_i —and we do not know their rates of speciation and extinction. However, we can model our uncertainty in p_i by letting it be a random variable drawn from a beta distribution. Allowing p_i to be a random variable helps statistically account for the confounding effects of variable diversification rate and lineage age on lineage richness (McPeck and Brown 2007, Scholl and Wiens 2016, Yu and Wiens 2024). Drawing p_i from a beta distribution averages over these possible confounding effects.

The resulting beta-geometric distribution has parameters α and β :

where B is the beta function.

Interestingly enough, the Yule-Simons distribution, historically developed to model number of species in genera (Yule 1925), is a special case of the beta-geometric distribution when $\beta = 1$ (Irwin 1975). In our beta-geometric model we will not fix $\beta = 1$ but instead allow it to be a free parameter, thus not making the strict birth-only assumption of Yule (1925).

Our final, quadratic model relating genus richness to average per-species abundance in a genus is

where x_i is average per-species abundance of lineage i . This model means that we allow the α parameter of the beta-geometric distribution to change with average per-species abundance but we assume the β parameter is fixed across lineages. A fixed β across lineages still allows

sufficient flexibility of the model without introducing additional parameters for which there may be insufficient power to reliably estimate. From this model our estimate for the observable diversification probability for lineage i is $1 - \hat{p}_i = 1 - \frac{\alpha_i}{\alpha_i + \beta}$.

To evaluate the significance of an intermediate peak in diversification across average per-species abundances (i.e. the quadratic term) we also fit a model without quadratic terms:

We then compare quadratic and linear models with a likelihood ratio test to evaluate significance of the intermediate peak.

While the *VGAM* package (Yee 2025) in R enables beta-geometric generalized linear modeling, its implementation is based on iterative Fisher scoring which is unnecessary and computationally slow for our application. Therefore, we use functionality from *VGAM* to write a custom likelihood maximization routine that we use to estimate model parameters. More details on this approach are provided in Supplementary Section S5.

Results

Simulating a birth-death-immigration process with speciation

1,464 out of 5,000 simulation runs (29.28%) resulted in speciation. Using a window size of 100 time steps resulted in 389,945 non-overlapping windows. Of those, 1,464 contained a speciation event. Thus the overall probability of speciation across windows is 0.004.

Figure 2 shows, as expected, that abundance is larger when $\lambda > \mu$ and smaller when $\lambda < \mu$, maintained by immigration. When $\lambda > \mu$ populations will grow indefinitely. When birth and death rates are equal ($\lambda = \mu$) the population could crash or expand depending on small random fluctuations and random immigration. In Figure 2 smaller abundances for $\lambda > \mu$ likely correspond to time bin windows earlier in the simulation as well as those that stochastically drifted to smaller sizes.

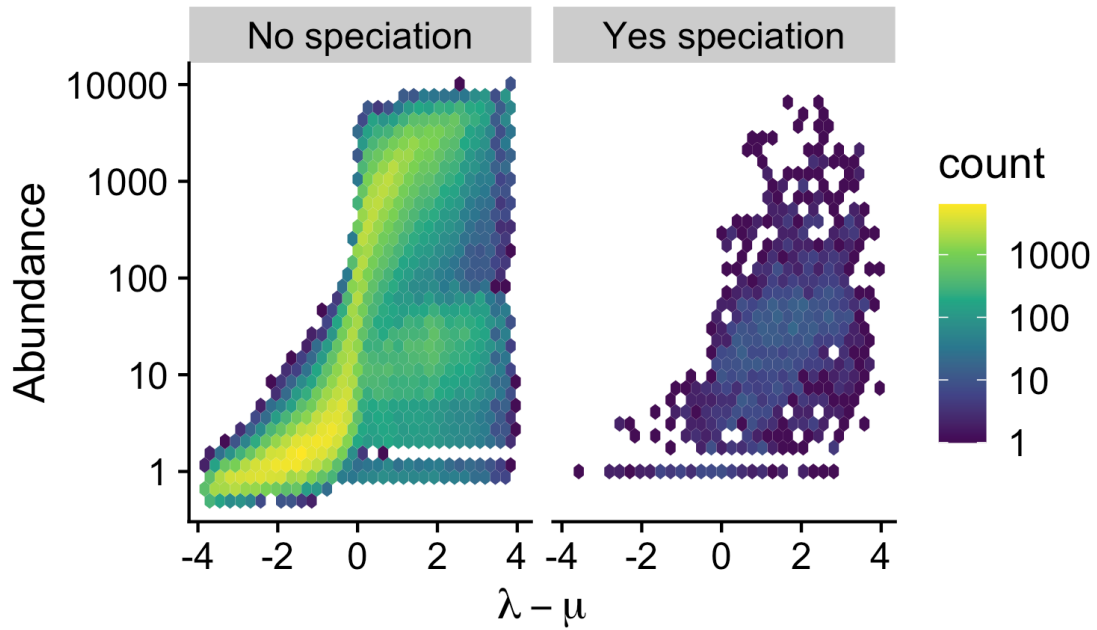


Figure 2: Relationship across simulation run windows between growth rate ($\lambda - \mu$) and abundance. Panels show simulation runs without and with full speciation. Colored hexagons show the number of points with values in that small neighborhood in order to more clearly show the distribution of simulation results.

Figure 3 shows how overall dispersal rate (global plus local) influences abundance. Only when growth rate is low does dispersal rate increase abundance, consistent with small abundances being maintained by immigration via a rescue-effect-like mechanisms (Brown and Kodric-Brown 1977).

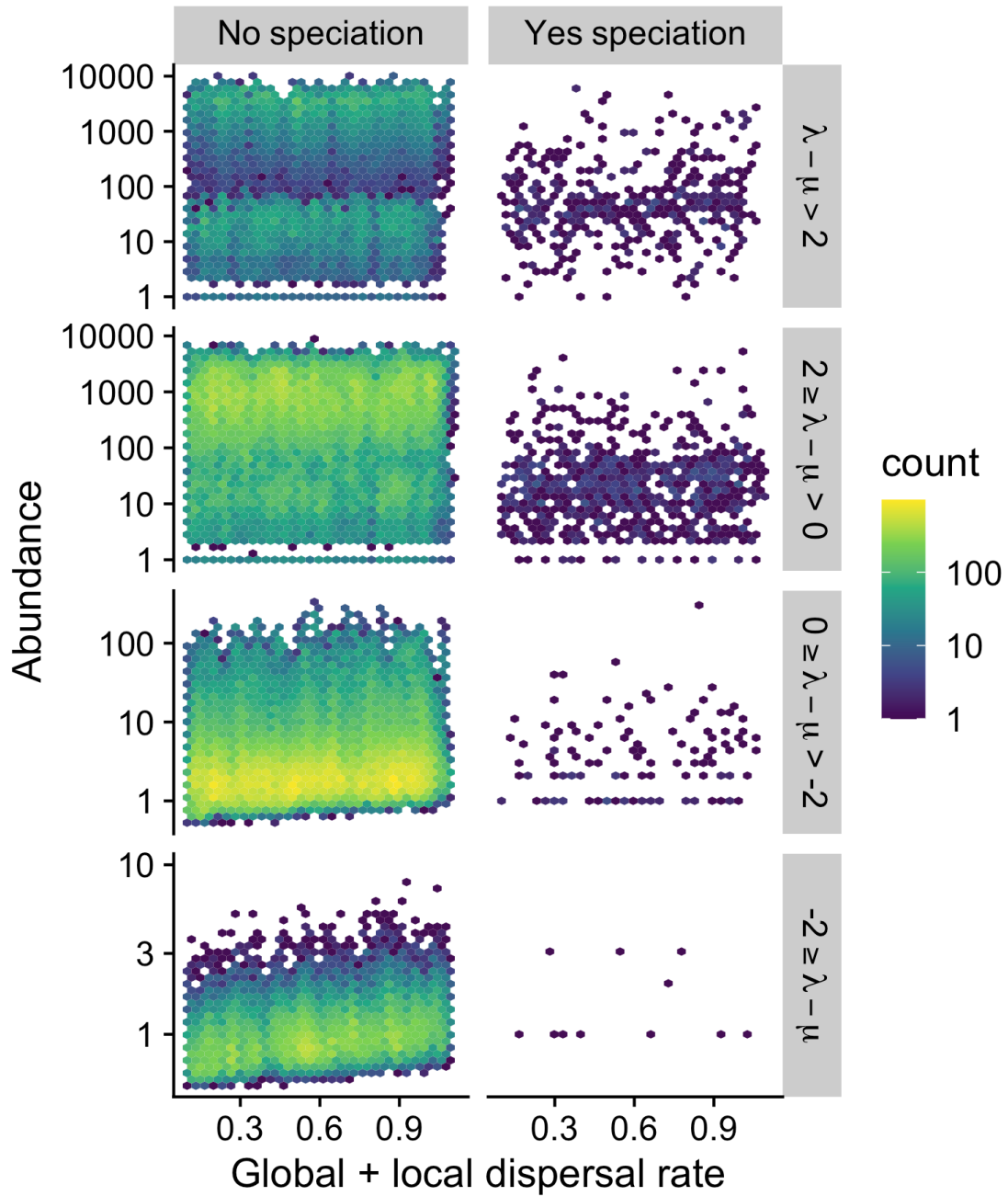


Figure 3: Relationship across simulation run windows between combined dispersal rate (global plus local) and abundance. Panel columns show simulation run windows without and with full speciation while panel rows show simulation run windows of different growth rate ($\lambda - \mu$). Colored hexagons show the number of points with values in that small neighborhood in order to more clearly show the distribution of simulation results.

339 Effect of abundance on speciation

340 Figure 4 shows that probability of speciation across simulation run windows peaks at
 341 intermediate abundance. A logistic mixed effects model with a random effect for simulation run
 342 and a quadratic term for abundance confirms this peak (95% CI of quadratic term is [-0.311, -
 343 0.257]). The estimated location of the peak in speciation probability is 105.022 with a 95% CI of
 344 [88.853, 124.134]

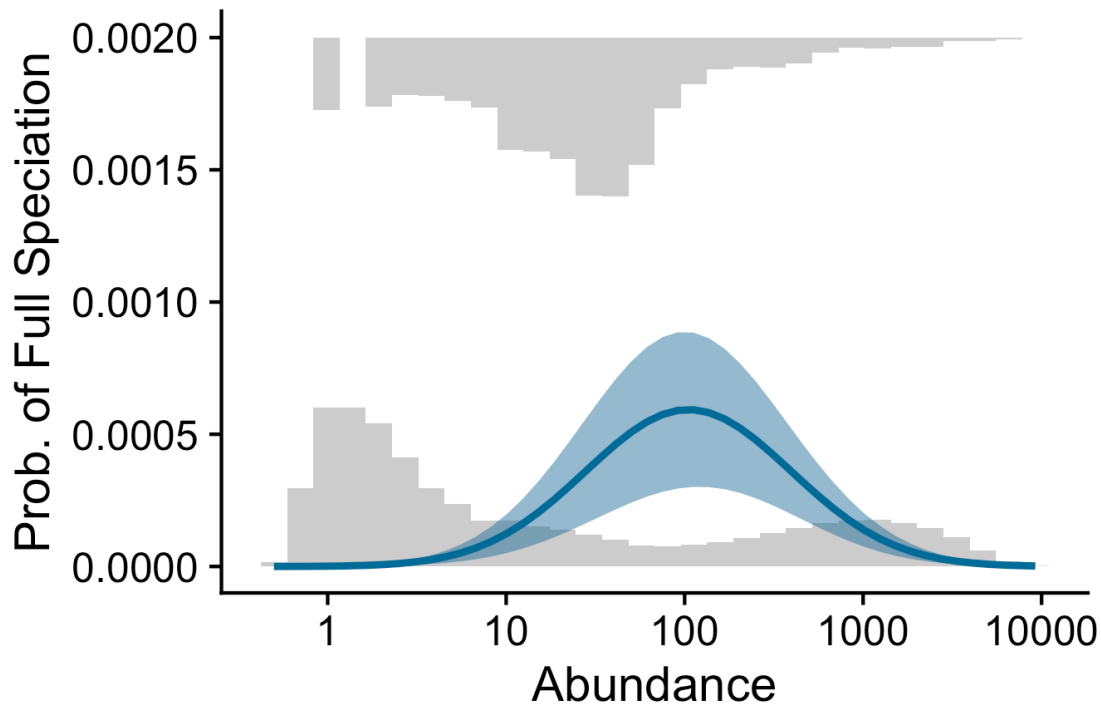


Figure 4: Relationship across simulation run windows between average abundance and speciation across. Histograms across bottom and top show the frequencies of windows without speciation (bottom) and with full-speciation (top) across different different incipient speciation rates. Curve is a binomial generalized linear model and shaded region shows 95% confidence envelope.

345 Effects of simulation model parameters on probability of speciation

346 Figure 5 shows the estimated effects on speciation probability from GLMMs that include
 347 simulation model parameters interacting with abundance. After accounting for the hump-shaped
 348 effects of abundance on probability of speciation, incipient speciation rate ν and growth rate $\lambda - \mu$
 349 both have positive impacts on probability of speciation. Conversely, after accounting for
 350 abundance, τ (the base wait time to speciation) and combined dispersal rate (local plus global)
 351 have no significant impact on probability of speciation. Across all GLMMs, the 95% CI of the
 352 quadratic term for abundance never crosses zero.

353 Figure 5 also shows how these GLMMs with simulation model covariates lead to different
 354 estimates of peak speciation probability. Relative to the GLMM with only a hump-shaped

relationship with abundance, incipient speciation rate and growth rate both shift the location of peak speciation probability lower. Greater dispersal shifts the location of the peak higher while τ has only a negligible effect.

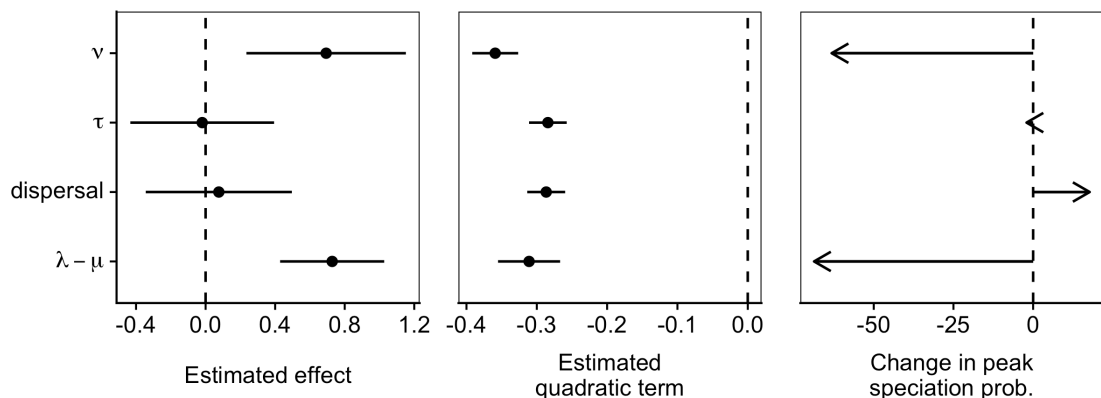


Figure 5: Summary of different GLMMs predicting speciation probability. Different models are shown across the y-axis, labeled by the simulation model parameter included as a covariate along with the hump-shaped relationship effect of abundance. Estimated effect sizes and 95% confidence intervals are shown in the left hand panel. The quadratic term for abundance from each model is shown in the middle panel. The right hand panel shows the shift in abundance value at which probability of speciation reaches its peak. This shift is shown relative to the model without simulation model parameters as covariates. The magnitudes each arrow represent how the location of peak speciation probability would shift with an increase in the simulation model covariate.

Relationship between genus abundance and species richness in Hawai'i arthropods

Arthropods in Hawai'i show a hump-shaped relationship between abundance and species richness (Figure 6). There is a clear outlier with higher genus species richness (above 300) than expected based on abundance. This outlier is *Drosophila*, which is not surprising given both the extensive systematics work on Hawaiian *Drosophila* and their unique speciation mechanisms (O'Grady and DeSalle 2018), both of which could inflate the group's reported species richness.

The hump-shaped relationship is captured by a quadratic generalized beta-geometric model with sample size $n = 118$ and coefficients *intercept* = 0.920 (95% CI: [0.207, 1.634]), *linear term* = -0.545 (95% CI: [-0.922, -0.168]), *quadratic term* = 0.100 (95% CI: [0.030, 0.170]). 95% CIs are Wald confidence intervals. Coefficient values are reported on the link scale (i.e. as in eq.). The location of maximum predicted genus richness (and therefore $1 - \hat{p}$) is 15.273 with 95% CI [8.641, 26.995].

A likelihood ratio test comparing quadratic and linear models of genus richness across average per-species abundance favors the quadratic model with a likelihood ratio of 7.012 and a P -value of 0.0081 from a χ^2 distribution with $d.f. = 1$. Removing *Drosophila* has no meaningful impact on the relationship or the statistical significance of the quadratic term (see Supplementary Section S5).

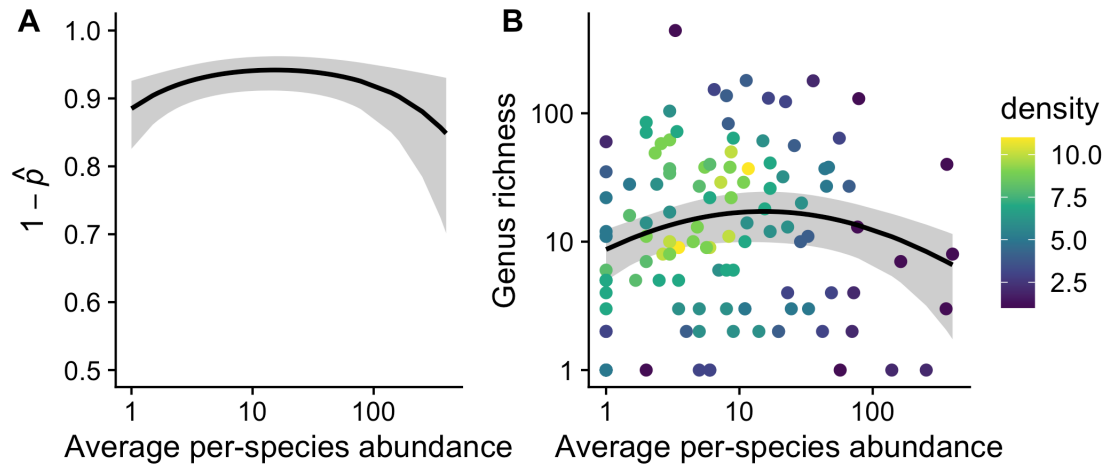


Figure 6: Relationship between genus abundance and species richness for arthropods endemic to Hawai'i. Panel A shows the fitted relationship between abundance and the observable diversification probability $1 - \hat{p}$. Solid line is expected value and gray shaded region is 95% confidence envelope. Panel B shows the pattern in the raw data along with the relationship predicted by the quadratic model. The solid black line represents the expected value and gray shaded regions the 95% confidence envelope. Colors of points reflect the local density of data points such that a yellow point indicates a cluster of greater than 10 data points at that specific x-y location and dark a purple point indicates only a single data point. Points coded by density were used instead of hexagons coded by count (as in Figures 2 and 3) because of the sparsity of these real data compared to the abundance of simulated data. The outlier with genus richness above 300 is *Drosophila* discussed further in the main text and Supplementary Section S5.

Discussion

Taken all together these results indicate support, from both theoretical and empirical perspectives, for the idea that intermediate abundances promote speciation. Using a theoretical simulation-based model we show that the chance of speciation peaks at intermediate abundances and that this peak is robust even when considering the interactive effects other model parameters might have on speciation. This result is consistent with the idea that these intermediate abundances represent a balance between dispersal, which slows speciation, and population persistence plus per capita incipient speciation, which are necessary for speciation. The balance would arise because total dispersal is lowest at low abundance (meaning the rate of transition from incipient to full speciation is greatest at low abundance) while persistence and incipient speciation are highest at high abundance. Our work suggests an intriguing possibility that if properties like abundance and dispersal are emergent species-level traits that modulate speciation, they could be acted upon by macroevolutionary selection (Rabosky and McCune 2010).

Our results are novel in that they go against the widespread assumption that more abundant lineages should be those that produce more new speciation—an assumption tracing back to Darwin (1859) and also integral to the unified neutral theory of biodiversity (Hubbell 2001). However, our theoretical results are also grounded in published hypotheses, specifically the role

of total dispersal in modulating speciation (Price and Wagner 2004, Claramunt et al. 2012, Czekanski-Moir and Rundell 2019, Yamaguchi 2022).

We arrive at these results by building on established modeling frameworks (Hubbell 2001, Rosindell et al. 2010), but we explore how abundance drives total dispersal, in turn swamping regional differentiation (Price and Wagner 2004, Birand et al. 2012, Claramunt et al. 2012, Czekanski-Moir and Rundell 2019, Ashby et al. 2020, Ciccheto et al. 2024). The model produces realistic patterns of abundance as responding to growth rate and, when growth rate is low, responding to dispersal via a rescue-effect like mechanism (Brown and Kodric-Brown 1977). The scaling of abundance and total dispersal is driven in large part by abundance (because abundances can reach large sizes) such that after accounting for abundance there is no residual effect on speciation probability of per capita dispersal, or of the base-level waiting time to full speciation in the absence of immigration. However, the GLMM including per capita dispersal, despite a seemingly non-significant dispersal term, does shift location of the peak probability of speciation. The shift is toward larger abundance. This would make sense if greater per capita dispersal slows (but does not prevent) eventual full speciation and the longer wait time to full speciation allows populations to reach a higher abundance.

The fact that greater incipient speciation rate increases the chance of full speciation is not surprising (more incipient events yield more random trials for possible full speciation). Similarly, the shift in peak speciation toward lower abundance with greater incipient speciation rate is likely due to a shorter total wait time to full speciation because incipient speciation happens more quickly. It might seem counter-intuitive that greater growth rate would decrease the abundance at which speciation peaks, but when growth rates are greater, then abundance is gained locally (increasing regional differentiation) without the need for immigration to buoy it; thus if full speciation is reached, it will be at lower abundances.

Our empirical results are qualitatively consistent with the theoretical finding that speciation peaks at intermediate abundances. The empirical results are made possible by the unique evolutionary setting of the Hawaiian Archipelago, where extreme isolation aids in the direct observation of diversification (Wagner and Funk 1995, Price and Wagner 2004). However, inferring past diversification from modern-only data has been shown to have multiple technical pitfalls including artifacts of sampling (Gruner et al. 2008), model non-identifiability (Louca and Pennell 2020), and difficulty in disambiguating effects of clade age and clade diversification rate (Yu and Wiens 2024). We test for sampling artifacts due to possibly under-sampling rare taxa (Supplementary Section S5.4) and due to choice of summary statistic (Supplementary Section S5.5) and find our results are robust to these checks.

The empirical and theoretical results are also distinct and cannot be interpreted as quantitatively equivalent. While both produce a hump-shaped relationship between abundance and measures of speciation or diversification, this does not mean that the simulation model parameters have a one-to-one mapping onto the parameters of a hypothetical diversification model that could have produced the empirical patterns. The parameters of the simulation model have to be interpreted separately from the empirical results because of key differences between the simulation set-up and the empirical analysis. Firstly, the “response variables” of each analysis are different—the simulation is able to directly observe speciation and thus we directly analyze speciation probability; conversely, we cannot directly observe speciation in the empirical system and so instead model $1 - \hat{p}$, the “observable diversification probability.” The fact that we stop a

simulation run once speciation is achieved also makes the simulated results different from the empirical system—in the real world, time does not stop if speciation occurs; instead, some clades continue to diversify and some go extinct. Our simulation model is thus one of the probability of first speciation and does not account for how the probability of speciation may itself change over the course of diversification (e.g., Rabosky 2009, Wagner et al. 2014, Calcagno et al. 2017, Foote 2023). The continued diversification of real world clades could also explain the flatter relationship of $1 - \hat{p}$ across abundances. Additionally, while both the simulation model and empirical analysis are concerned with abundance, the way abundance is recorded in each is different. In the simulation we know the complete abundance of every local population and we scale rates such that local abundances are computationally tractable. Real world abundances (perhaps up to 10^8) would not be computationally manageable. Also unlike the simulations, the real world data represent tiny subsamples of the whole populations.

Nonetheless, the simulation model provides a proof-of-concept and a mechanism for how speciation might depend on abundance in a more nuanced way than previously considered. Critically, our model does not depend on a neutrality assumption to arrive at this conclusion. Although our model is inspired by the agent-based modeling approaches of neutral theories (Hubbell 2001, Rosindell and Harmon 2013, Rosindell et al. 2015) we only model populations of a single species and thus make no assumption about how multiple species interact or whether different species have different eco-evolutionary rates. Thus our findings apply regardless of whether one assumes neutral- or niche-based dynamics. We also do not assume density dependence, as this is not our focus. However, one can postulate that positive density-dependent dispersal (Travis et al. 1999), which causes total dispersal rate to increase super-linearly with abundance, could further accentuate our findings. Conversely, negative density-dependent dispersal (Rodrigues and Johnstone 2014) could dampen the effects we find, but, so long as per-capita dispersal remains a non-zero probability, likely not eliminate them.

Because our model focuses on a single species and thus makes no assumptions about dynamics or interactions between species we highlight the unnecessary assumptions of previous approaches. Darwin (1859), for example, was also interested in the relationship between abundance and speciation. He assumed that more abundant species gave rise to more new species because those abundant species were biologically dominant to less abundant one, and that this dominance promoted both abundance and diversification (Darwin 1859, Maurer 1999). The language of Darwin (1859), including reference to “conquer[ing]” [p. 125] and “triumphant” [p. 126] clades, cannot be separated from discourse then and now that seeks to justify racism and colonial violence (Brantlinger 2003, Hesse 2007, Fuentes 2021, Diogo et al. 2023). Beyond the inherent amorality of such assumptions, our work shows they are unnecessary in explaining why abundance connects to speciation. We also show that these unnecessary assumptions lead to a prediction—a positive relationship between abundance and speciation—that may not universally hold.

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